DeepXDE:
A deep learning library for solving differential equations

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DeepXDE: Capabilities

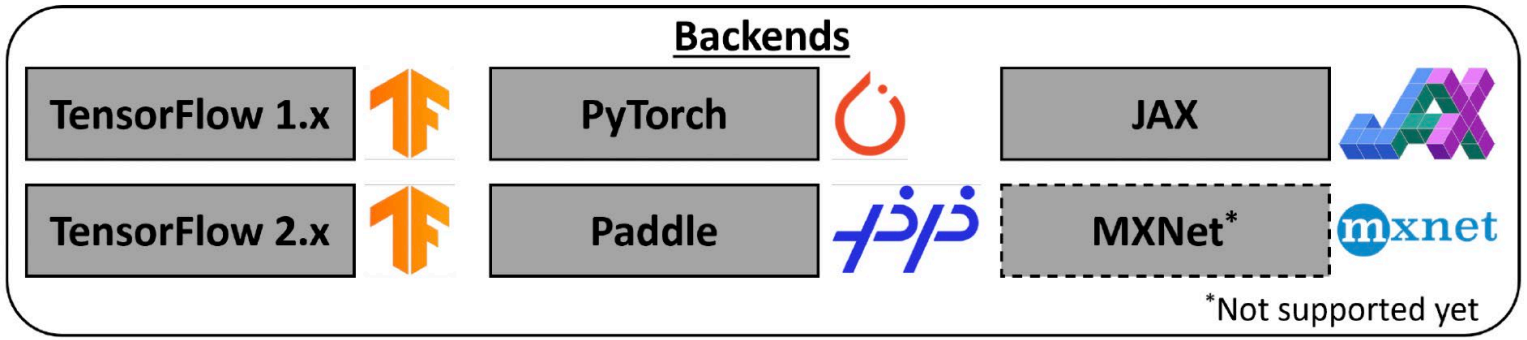
- Physics-Informed Neural Network (PINN)
  - solving forward/inverse ODEs/PDEs
  - solving forward/inverse integro-differential equations (IDEs)
  - fPINN: solving forward/inverse fractional PDEs (fPDEs)

- Deep Operator Network (DeepONet)
  - learning operators
  - MIONet: learning multiple-input operators
  - DeepM&Mnet: solving multiphysics and multiscale problems

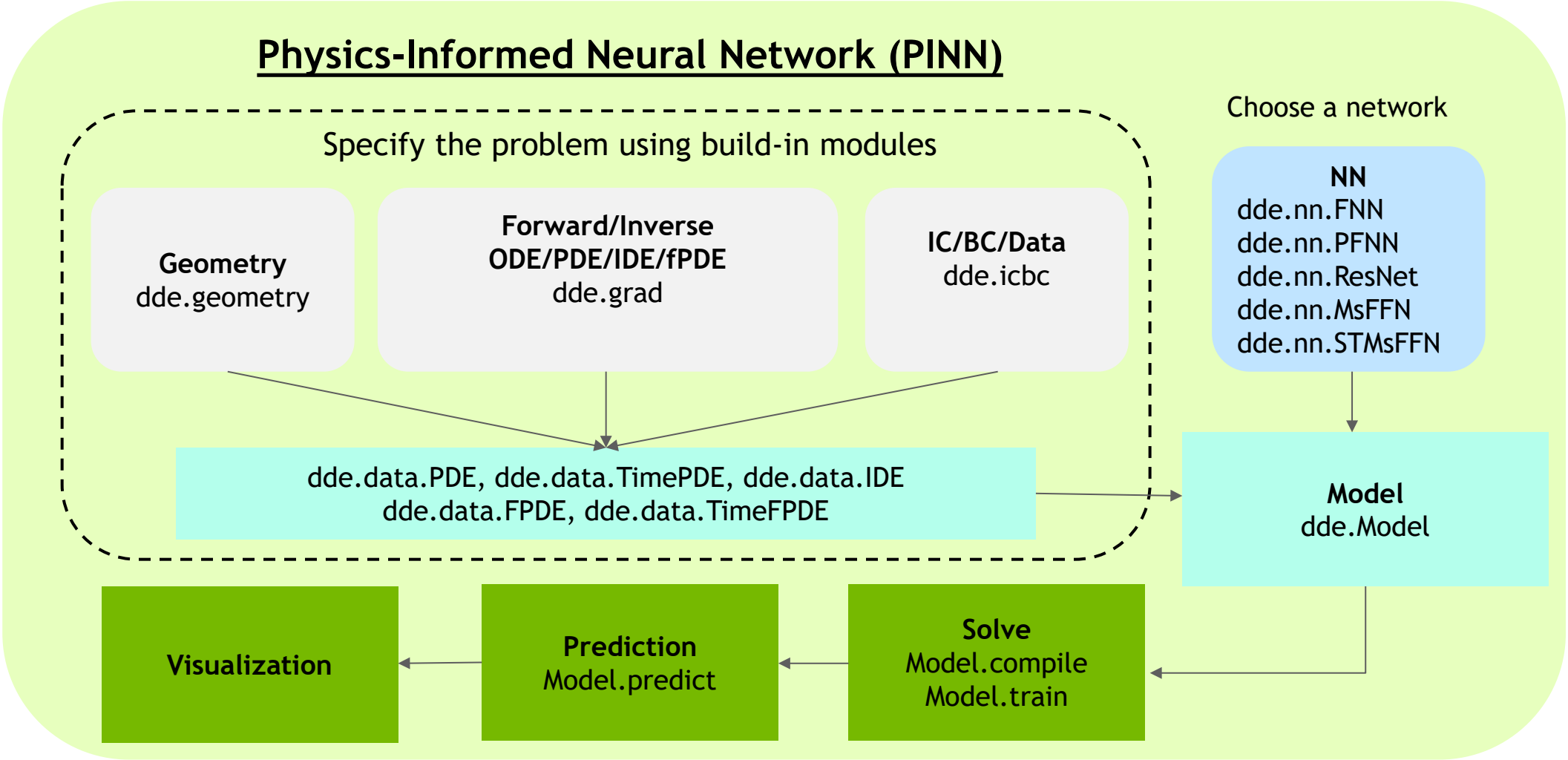
- Multifidelity Neural Network (MFNN)
  - learning from multifidelity data

- Lu, Meng, Mao, Karniadakis. DeepXDE: A deep learning library for solving differential equations. *SIAM Review*, 2021.

GitHub: https://github.com/lululxvi/deepxde
Documentation: https://deepxde.readthedocs.io



Workflow for PINNs using DeepXDE



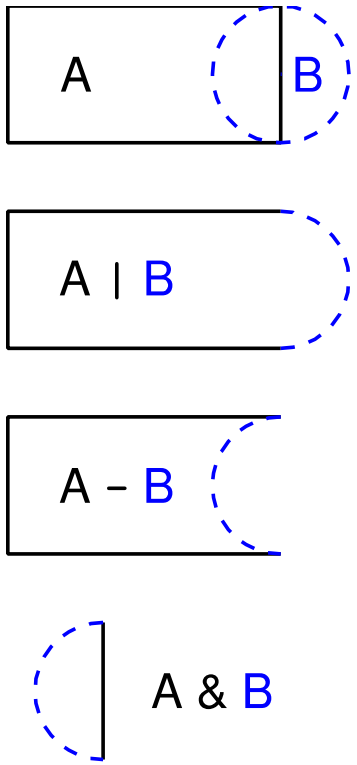
Workflow for PINNs using DeepXDE

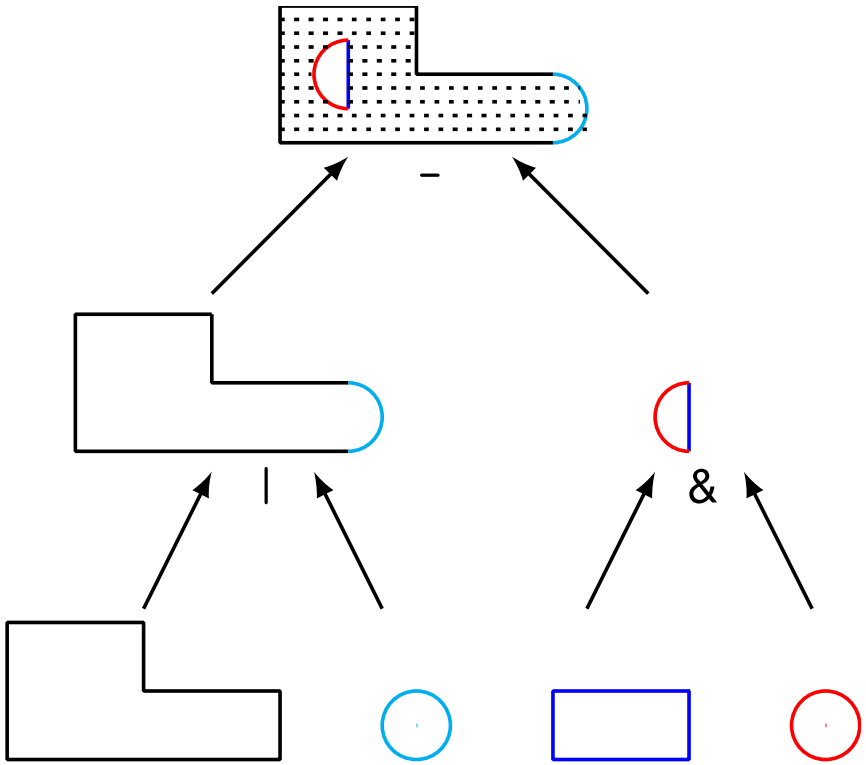
1. Specify the computational domain using the `geometry` module.
2. Specify the PDE.
3. Specify the ICs and BCs.
4. Combine the geometry, PDE and ICs/BCs into `data.PDE` or `data.TimePDE` for time-independent or time-dependent problems.
5. Construct a neural network using the `nn` module.
6. Define a `Model` by combining the PDE problem in Step 4 and the neural net in Step 5.
7. Call `Model.compile` to set the optimization hyperparameters, such as optimizer and learning rate.
8. Call `Model.train` to train the network from random initialization or a pre-trained model.
9. Call `Model.predict` to predict the PDE solution at any location.

Complex domains using DeepXDE

- Primitive geometries
  - `interval`, `triangle`, `rectangle`, `polygon`, `disk`, `cuboid`, `sphere`
- boolean operations:
  - Union $A|B$
  - difference $A - B$
  - intersection $A\&B$

- $geom1 = dde.geometry.Rectangle([0,0],[2,1])$
- $geom2 = dde.geometry.Disk([2,0.5],0.5)$
- $geom_union = geom1 \mid geom2$
- $geom_diff = geom1 - geom2$
- $geom_inter = geom1 \ \& \ geom2$





Demo 1: 1D Poisson equation

- Governing Equation

$$-\frac{d^2u}{dx^2} = \pi^2 \sin(\pi x),$$

  where $x \in [-1, 1]$.

- Dirichlet boundary conditions are,

$$u(x = -1) = u(x = 1) = 0.$$

- Exact solution is,

$$u(x) = \sin(\pi x)$$

➢ **Goal is to obtain an approximate solution using PINNs**

Demo 2: Diffusion Equation

- Governing Equation

$$\frac{du}{dt} = \frac{d^2u}{dx^2} - \exp^{-t}(\sin(\pi x) - \pi^2 \sin(\pi x))$$

  where $x \in [-1, 1]$ and $t \in [0, 1]$.

- Initial condition is,

$$u(x, 0) = \sin(\pi x).$$

- Dirichlet boundary conditions are,

$$u(-1, t) = u(1, t) = 0.$$

- Exact solution is,

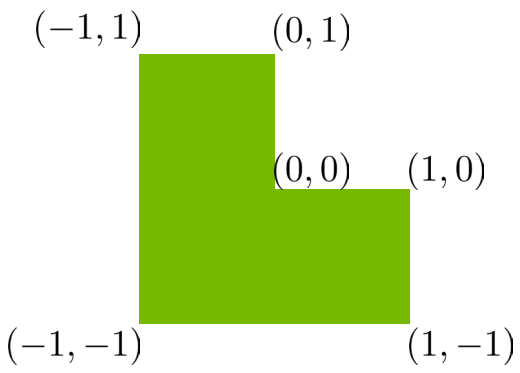
$$u(x, t) = \exp^{-t} \sin(\pi x).$$

➢ Goal is to obtain an approximate solution using PINNs

Demo 3: Poisson equation in L-shaped domain

- Governing Equation

$$-\frac{\partial^2 u}{\partial x^2} - \frac{\partial^2 u}{\partial y^2} = 1$$



- Dirichlet boundary conditions are,

$$u(x, y) = 0$$

where $(x, y) \in \Gamma$

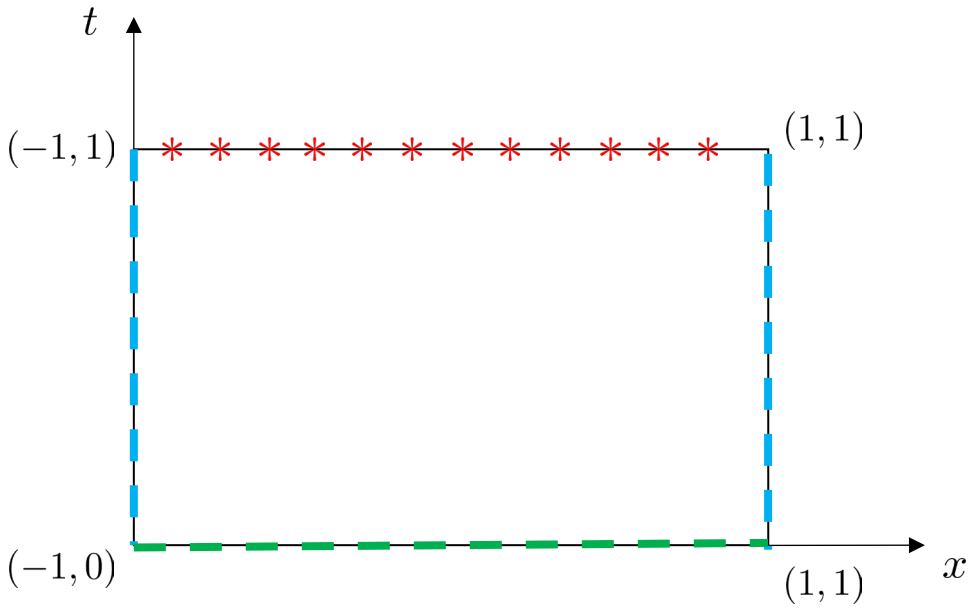
➢ **Goal is to obtain an approximate solution using PINNs**

Demo 4: Inverse problem

- Governing diffusion equation

$$\frac{\partial u}{\partial t} = \boxed{\mathbf{C}}\frac{\partial^2 u}{\partial x^2} - e^{-t}(\sin(\pi x) - \pi^2 \sin(\pi x))$$

- where $x \in [-1, 1]$, $t \in [0, 1]$
- Initial condition: $u(x, 0) = \sin(\pi x)$
- Dirichlet boundary condition: $u(-1, t) = u(1, t) = 0$



Exact Solution: $u(x, t) = e^{-t} \sin(\pi x)$

True $\boxed{\mathbf{C}} = 1$

➢ **Goal is to infer the parameter C and simultaneously obtain solution for U given data** $*$ — —